PatientTriage.ai: Business Proposal & Strategy

“Triage is a snapshot. Risk isn't.” — Transforming Emergency Safety & Throughput

Accenture Innovation Challenge 2026 • Round 2 Business Case & Phased Roadmap

NET ANNUAL ROI	AVOIDED ICU COLLAPSE	LWBS RECOVERY	TARGET MARKET
$3.82M / Hospital	**-64% Decompensation**	**+$1.12M Recovered**	5,500+ US/EU Emergency Depts

Executive Summary

Emergency department (ED) crowding is a global healthcare crisis with over 140M annual visits in the US alone. Traditional emergency triage operates on an outdated premise: **a single static snapshot taken at intake**. Once triaged, patients wait unmonitored for 3 to 6 hours. When patients silently deteriorate, the results are catastrophic: preventable in-hospital cardiac arrests, unanticipated ICU transfers, heightened malpractice claims, and severe nurse burnout.

PatientTriage.ai delivers a continuous physiological safety layer that monitors vital velocity (ΔSpO■, ΔHR), dynamic evidence decay ($\tau_{staleness}$), data uncertainty (*Unknown is NOT Safe*), and physician coverage (**The Attention Gap**). It generates **$3.82M in net annual value** per 500-bed hospital while eliminating silent waiting room mortality.

1. Problem Framing: 3 Systemic Failure Modes of Static Triage

- **1. Silent Waiting Room Decompensation:** ESI Level 3/4 patients silently worsen without automated alerts until physical collapse.
- **2. The 'Missing Data Is Safe' Assumption:** Missing vitals default to low urgency. In acute medicine, *Unknown is NOT Safe*.
- **3. The Attention Bottleneck:** Attended critical cases block static lists, while *unattended deteriorating patients* remain buried.

2. Solution Design & 3-Tier Layered Architecture

PatientTriage.ai enforces a strict 3-tier architecture: **Tier 1: Deterministic Safety Layer** (hard red-flags and downgrade blocking); **Tier 2: AI Decision-Support Layer** (vital velocity, dynamic confidence decay, and Attention Gap re-ranking); and **Tier 3: Clinician Governance Layer** (licensed clinician override with append-only audit logging). Runs air-gapped on edge hardware with sub-15ms latency.

3. Target Users & Stakeholder Value Propositions

Stakeholder	Primary Clinical / Financial Pain Point	PatientTriage.ai Value Proposition
Triage Nurses (RNs)	Overwhelmed tracking 40+ waiting patients; fear of silent deterioration.	Surfaces Top 3 actionable tasks with single Next-Best-Action buttons.
Emergency MDs	Blind to which waiting patient has worsened since initial intake.	Attention Gap Queue dispatches doctors to highest unmet clinical need.
Chief Medical Officers	Delayed diagnosis lawsuits, sentinel events in waiting lounges.	100% Downgrade Guardrails & append-only audit trail for malpractice defense.
Hospital CFOs	Uncompensated ICU boarding, LWBS revenue leakage, nurse turnover.	Delivers measurable $3.82M annual net ROI per 500-bed hospital facility.

4. Business Case, Financial ROI & Impact Model (500-Bed Hospital)

Model based on 65,000 annual ED visits, 500 acute care beds, and $1,200 average ED revenue per visit:

Value Driver	Pre-Implementation Baseline	Post-Implementation Impact	Annual Financial Value
1. LWBS Revenue Recovery	3,120 patients/yr leave (4.8%)	30% reduction via proactive re-engagement	**+$1,123,200 / yr**
2. Avoided ICU Transfers	145 waiting room ICU crashes/yr	64% reduction (93 avoided ICU stays @ $15k)	**+$1,395,000 / yr**
3. Malpractice Risk Mitigation	$1.2M annual liability reserve	40% reduction via documented audit trail	**+$480,000 / yr**
4. Nurse Retention & Overtime	26.8% nurse turnover (14 replacements)	4 replacements avoided + 15% overtime reduction	**+$378,000 / yr**
5. ED Throughput & Boarding	248 mins average wait/boarding	30-minute reduction via optimized dispatch	**+$445,000 / yr**
TOTAL GROSS ANNUAL VALUE	—	—	**$3,821,200 / yr**
Software Subscription & Support	—	Enterprise Tier License	-$240,000 / yr
NET ANNUAL ROI TO HOSPITAL	—	**14.9x Net Return on Investment**	**+$3,581,200 / yr**

5. Commercialization, Pricing & Accenture Synergy

- **Tiered B2B SaaS Subscriptions:** Community Rural Clinic ($48k/yr), Regional Acute Hospital ($120k/yr), Academic Level-1 Trauma Center ($240k/yr).
- **One-Time Integration Services:** $75,000–$150,000 covering HL7 FHIR connector setup and clinician workflow onboarding.
- **Accenture Strategic Synergy:** Accenture Health & Life Sciences serves as the prime system integrator, bundling PatientTriage.ai into broader Emergency Department Digital Transformation and Throughput Optimization engagements.

6. Phased Multi-Year Implementation Roadmap

Phase	Timeline	Milestones & Deliverables
Phase 1: Lab Validation	Q3 2026	20 benchmark scenarios validated across 33 automated tests; sub-15ms inference latency.
Phase 2: Shadow Trial	Q4 2026	Silent shadow deployment alongside Epic/Cerner via HL7 FHIR; clinician concordance evaluation.
Phase 3: Live Hospital Pilot	Q1–Q2 2027	Live clinical decision-support at 2 partner health systems; track LWBS & ICU avoidance KPIs.
Phase 4: Enterprise Scale	Q3 2027+	Regional hospital network rollout with multi-facility dashboarding and telemedicine integration.

7. Key Risks, Regulatory Compliance & Mitigations

- **AI Hallucination Risk:** Mitigated by 3-tier architecture where deterministic safety red-flags override all statistical models.
- **Clinician Alarm Fatigue:** Mitigated by queue compression surfacing only the top 3 actionable tasks rather than flooding nurses.
- **Regulatory Classification (SaMD):** Positioned as Non-Device CDS (21 U.S.C. § 360aaa-1); clinician retains 100% decision authority.
- **Data Privacy (HIPAA/GDPR):** Runs air-gapped on local hospital network with zero external cloud LLM dependencies and append-only audit logging.

Conclusion: PatientTriage.ai delivers the technological breakthrough hospitals urgently require: an intelligent, explainable, and ethically grounded continuous safety layer that protects patients, empowers clinicians, and unlocks millions in enterprise hospital value.